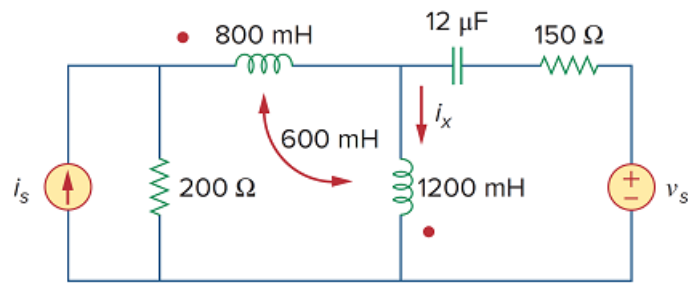


Problem

Use mesh analysis to find i_x in Fig. 13.80, where $i_s = 4 \cos(600t)$ A and $v_s = 110 \cos(600t + 30^\circ)$

Figure 13.80



Step-by-step solution

Step 1 of 9

Convert the expressions of voltage and currents into phasor form.

$$i_s = 4 \cos(600t) \text{ A}$$

$$\mathbf{I}_s = 4 \angle 0^\circ$$

$$v_s = 110 \cos(600t + 30^\circ) \text{ A}$$

$$\mathbf{V}_s = 110 \angle 30^\circ$$

The value of frequency is, $\omega = 600 \text{ rad/s}$.

Step 2 of 9

Calculate the impedance of inductors.

$$\begin{aligned} \mathbf{Z}_{L_1} &= j\omega L_1 \\ &= j(600)(800 \times 10^{-3}) \\ &= j480 \Omega \end{aligned}$$

$$\begin{aligned} \mathbf{Z}_{L_2} &= j\omega L_2 \\ &= j(600)(1200 \times 10^{-3}) \\ &= j720 \Omega \end{aligned}$$

Step 3 of 9

Calculate the impedance of capacitor.

$$\begin{aligned} \mathbf{Z}_C &= \frac{1}{j\omega C} \\ &= \frac{1}{j(600)(12 \times 10^{-6})} \\ &= -j138.89 \Omega \end{aligned}$$

Step 4 of 9

Draw the following circuit diagram in frequency domain.

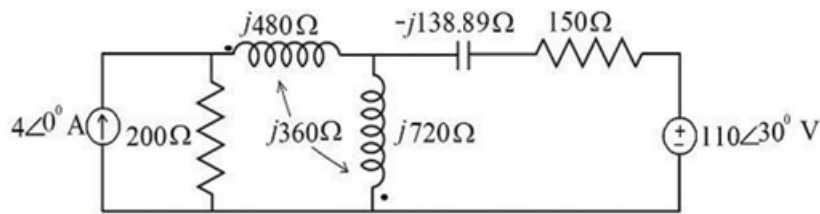


Figure 1

Step 5 of 9

Use source transformation technique, convert current source into the voltage source.

$$\begin{aligned} V &= IR \\ &= (4\angle 0^\circ)(200) \\ &= 800\angle 0^\circ \end{aligned}$$

Draw the modified circuit diagram as shown in Figure 2.

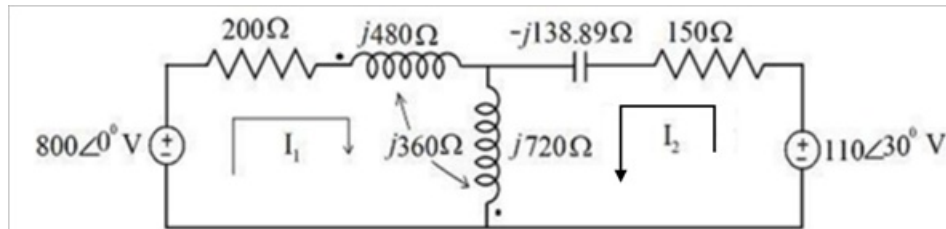


Figure 2

Step 6 of 9

Use the method of dot convention to determine the polarity of the mutual voltage.

Since the current I_1 is entering the inductor L_1 , the polarity of the mutual voltage at the dot of the inductor L_2 is positive.

The mutual voltage is,

$$\begin{aligned} v_{m2} &= j\omega M I_1 \\ &= j(600)(600 \times 10^{-3}) I_1 \\ &= j360 I_1 \end{aligned}$$

Since the current $(I_1 + I_2)$ is leaving the inductor L_2 , the polarity of the mutual voltage at the dot of the inductor L_1 is negative.

The mutual voltage is,

$$\begin{aligned} v_{m2} &= j\omega M (I_1 + I_2) \\ &= j(600)(600 \times 10^{-3}) (I_1 + I_2) \\ &= j360 (I_1 + I_2) \end{aligned}$$

Replace the effect of mutual voltages with a voltage source v_{m1} and v_{m2} in series with the respective inductor.

Note v_{m1} and v_{m2} are voltage drop in the mutual inductance and are not the voltage across the inductor. Figure 3 only shows the polarity of these mutual voltages in the inductor.

Step 7 of 9

Sketch the modified circuit as shown in Figure 3.

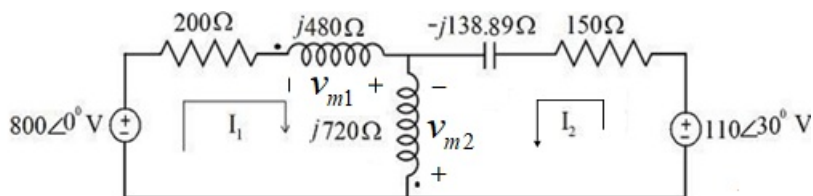


Figure 3

Apply Kirchhoff's voltage law in Loop 1.

$$\begin{aligned}
 -800\angle 0^\circ + (200 + j480)I_1 + (j720)(I_1 + I_2) - v_{m1} - v_{m2} &= 0 \\
 -800\angle 0^\circ + (200 + j480)I_1 + (j720)(I_1 + I_2) - (j360)(I_1 + I_2) - j360I_1 &= 0 \\
 (200 + j480)I_1 + (j360)I_2 &= 800\angle 0^\circ \dots\dots (1)
 \end{aligned}$$

Step 8 of 9

Apply Kirchhoff's voltage law in Loop 2.

$$\begin{aligned}
 110\angle 30^\circ &= (150 - j138.39)I_2 + j720(I_1 + I_2) - v_{m2} \\
 110\angle 30^\circ &= (150 - j138.39)I_2 + j720(I_1 + I_2) - j360I_1 \\
 110\angle 30^\circ &= j360I_1 + (150 + j581.61)I_2 \\
 j360I_1 + (150 + j581.61)I_2 &= 95.26 + j55 \dots\dots (2)
 \end{aligned}$$

Step 9 of 9

Solve equations (1) and (2).

$$\begin{aligned}
 I_1 &= 1.296 - j1.564 \text{ A} \\
 I_2 &= -0.859 + j0.5835 \text{ A}
 \end{aligned}$$

Determine the value of current i_x .

$$\begin{aligned}
 i_x &= I_1 + I_2 \\
 i_x &= 1.296 - j1.564 - 0.859 + j0.5835 \\
 i_x &= 0.437 - j0.9808 \text{ A} \\
 i_x &= 1.073\angle -65.98^\circ \text{ A}
 \end{aligned}$$

In sinusoidal form,

$$i_x = 1.073 \cos(600t - 65.98^\circ)$$

Therefore, the expression of current i_x is $1.073 \cos(600t - 65.98^\circ) \text{ A}$.